

Future-proof your data!

Making data FAIR

Making your data accessible and usable in the future means making your data FAIR. Adopting the FAIR principles makes it easier for others to find and reuse your data, increasing potential collaboration opportunities and ensuring acknowledgement of your data in other publications.

FAIR is a structured framework to manage the information about your research collection and ensure it is *findable, accessible, interoperable, and reusable* (F.A.I.R.) in the long-term.

The FAIR principles are designed to make it easier for humans and machines to locate, access and use your data, while promoting long-term data impact and preservation.

Making your data FAIR does not mean it has to be open, as some data is sensitive or subject to privacy and/or security restrictions. FAIR means that even if restrictions apply to the data, the conditions governing access are clear and transparent.

Let's take a closer look at the four pillars of the FAIR principles.



Figure 1 <https://openscience.eu/article/infrastructure/guide-fair-principles>

Findable

This means that the data has enough information (metadata) to be discoverable by humans and machines. This includes:

- Assigning the data a persistent identifier (eg. DOI or URL)
- Ensuring that the information about the data (metadata) is sufficiently descriptive
- Indexing the data in searchable resources

Accessible

This means that, once found:

- The data is retrievable using a standard communication protocol (eg. HTTP, HTTPS, FTP)
- There is a well-documented path for people and machines to download the data
- If the data is sensitive, there is a functioning authentication system to control access
- The information describing the data (metadata) is accessible even if the data is not

Interoperable

This means that the data can be easily integrated with other data and applications for analysis, storage and processing. This includes:

- Using standard vocabularies or data dictionaries to describe your data
- Referencing and describing relationships to other data

Reusable

This means your data contains sufficient information to ensure it can be reused, including:

- Clear licencing and rights statements that are machine readable
- Information about the provenance of the data
- Use discipline-specific metadata schema to ensure it is accessible to other researchers in the domain

Assess how FAIR your data is

The ARDC provides an online [FAIR data assessment tool](#).

The tool asks these questions to help assess your data against the FAIR principles.

PRINCIPLE	QUESTION	COMMENTS
FINDABLE	Does the dataset have identifiers assigned?	DOI, PURL, handle, URL
	Is the dataset identifier included in all metadata records describing the data?	Does the citation of the data include the DOI/URL etc?
	How is the data described by the metadata record?	Machine-readable or text-based?
	What type of searchable repository or registry is the metadata record in?	Public? Local?
ACCESSIBLE	How accessible is the data?	Public? Private?
	Is there access to the data online? (pending access	Web? Download? No access?

	approval)	
	Will the metadata record remain accessible even if the data isn't?	
INTEROPERABLE	What file format is the data available in?	Structured? Unstructured? Machine readable?
	What types of vocabularies are used to describe data elements?	Published vocabularies, data dictionaries
	How is the relationship to other data described in the metadata?	Resource description framework, URI
REUSABLE	Are the licencing and usage rights included in the metadata?	Licence URL (eg. Creative commons)
	Does the information about the data include enough detail about provenance to facilitate reuse?	Standard schema? Machine-readable? Structured?

The Policy Context

Curated Collections is hosted on the Commonwealth National Collaborative Research Infrastructure Strategy (NCRIS) facility managed by the Australian Research Data Commons (ARDC).

NCRIS policy requires that:

data generated, created, captured or stored by NCRIS funded projects will be made available to the wider research community based on the F.A.I.R. principles.

The ARDC also requires the FAIR principles to be applied to its own materials and to outputs from co-investment projects.

References

This module has been developed using the following resources:

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Department of Education. *National Collaborative Research Infrastructure Strategy 2023 Guidelines*. 2023. Accessed May 7, 2025. <https://www.education.gov.au/national-research-infrastructure/resources/national-collaborative-research-infrastructure-strategy-2023-guidelines>

GO FAIR. (n.d.). FAIR Principles. <https://www.go-fair.org/fair-principles/>

Open Science EU. (n.d.). A Guide to the FAIR Principles: Best Practices for Data Handling. <https://openscience.eu/article/infrastructure/guide-fair-principles>

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